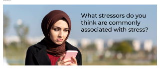
Stress and Mood Disorders

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We all experience **stress,** the physiological and psychological response to a threat or a challenge. Stress can be positive, but often it is negative, a state brought about by the feeling that we can't successfully cope with the demands being placed upon us. When stress is prolonged or chronic, it can affect how we think, feel, and act. Although some of these responses may allow us to cope with the stressor in the short term, in the long term (e.g., chronic stressors), they might lead to developing depression, bipolar disorder, or other psychological disorders (Aas et al., 2016; Hammen, 2005; Hart et al., 2021; Shin et al., 2017).

In fact, stress is a strong predictor of depression (Hammen, 2015). Stress has been found to influence both the onset and the persistence of depression (LeMoult, 2020). However, the relationship between stress and depression is complicated for two reasons. First, not everyone who is exposed to stressful life events develops depression. Second, just as stress leads to depression, the relationship is reciprocal with depression leading to stress (LeMoult, 2020).



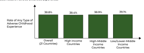
Research that has investigated the impact of particular stressors has found that a wide range of stressors are linked to mood disorders (Anderson et al., 2022; Hammen, 2005; Lippard & Nemeroff, 2019; Tennant, 2002). For example, research shows that adverse childhood experiences (Mansueto et al., 2021; Park et al., 2020), poverty (Heflin & Iceland, 2009), and experiencing discrimination due to race (Schulz et al., 2006), gender (Stepanikova et al., 2020), or sexual orientation (Sutter & Perrin, 2015) contribute to mood disorders.

Adverse Childhood Experiences

One type of stressor has been consistently linked to mood disorders—**adverse childhood experiences** (Lippard & Nemeroff, 2019; Mansueto et al., 2021), which are negative events experienced before the age of 18. Adverse childhood events are often separated into two categories: child abuse and neglect, and household challenges. Child abuse and neglect include physical abuse, emotional abuse, sexual abuse, physical neglect, and emotional neglect. Household challenges include exposure to the mother being treated violently, parental divorce or separation, parental incarceration, a household member with substance abuse problems, and a household member with mental illness. Researchers typically find that as the number of adverse childhood experiences a person experiences increases, so too, does the likelihood of their experiencing depressive symptoms in adulthood (Chapman et al., 2004).

How common are adverse childhood experiences? In a survey of over 200,000 adults in the United States, over 60% of participants reported experiencing an adverse childhood experience, including physical abuse, emotional abuse, sexual abuse, household mental illness, household substance use, incarcerated household member, parental separation or divorce, and household domestic violence (Merrick et al., 2018). In 2021, an estimated 600,000 children in the U.S. were victims of abuse and neglect (Child Welfare Information Gateway, 2023). On a global scale, a report from adults in 21 countries worldwide revealed that 38.8% of respondents reported any type of adverse childhood experience, with parental death being the most common (12.5%) (Kessler et al., 2010). A comparison of the reported adverse childhood experiences among countries based on income level is shown in the *Global Rates of Adverse Childhood Experiences* graph. Only minor differences were observed among the income level groups.

Global Rates of Adverse Childhood Experiences



Note. A bar graph illustrates the overall rate of adverse childhood experiences, as well as a comparison of rates among high income countries (Belgium, France, Germany, Israel, Italy, Japan, The Netherlands, Spain, USA), high-middle income countries (Brazil, Bulgaria, Lebanon, Mexico, Romania, South Africa), and low/lower-middle income countries (Colombia, India, Iraq, Nigeria, People's Republic of China, Ukraine).

Childhood Trauma and Depressive Disorders

Researcher Rosemay Remigio-Baker and her colleagues examined the association between adverse childhood experiences and current depressive symptoms in women in Hawaii (Remigio-Baker et al., 2014). They found household dysfunction, and physical, verbal, and sexual abuse experienced in childhood predicted depressive symptoms in adulthood, with verbal abuse having the strongest association with depressive symptoms. Another study examined the relationship between a standard set of adverse childhood experiences and depression and suicide attempts (Merrick et al., 2017). Compared with individuals did not experience adverse childhood experiences, those who experienced six or more different adverse childhood experiences were almost three times more likely to report depressive symptoms and 24 times more likely to attempt suicide (Merrick et al., 2017).

Childhood Trauma and Bipolar Disorders

Individuals with bipolar disorder report experiencing more stressful life events than healthy controls, and more stressful life events occurring before mood episodes than during symptom-free periods (Lex et al., 2017). Further, the death by suicide of a mother or sibling was found to be associated with an increased likelihood of hospitalization (for the for bipolar symptoms (Kessing et al., 2004). It is important to note that the relationship between stressful life events and bipolar disorder is somewhat complicated; research has demonstrated that stressful life events precede mood episodes but also that mood episodes precede stressful life events (Koenders et al., 2014).

Monica Aas and colleagues (2016) conducted a review of research on the role of childhood trauma in bipolar disorder. They concluded that all subcomponents of childhood trauma (emotional, physical, and sexual abuse) are strongly linked with bipolar disorder, yet there is continued debate on the role of each trauma subtype. Emotional abuse is the only subtype which may have dose-dependent links with bipolar disorder (Aas et al., 2016). Other studies have also found that emotional neglect in childhood is more for patients suffering from bipolar disorder. For example, Watson and colleagues (2014) compared adult outpatients with bipolar disorder with control participants on the Childhood Trauma Questionnaire. Patients with bipolar I disorder and bipolar II disorder experienced significantly more emotional neglect than control participants (Watson et al., 2014). Overall, childhood trauma has been consistently linked to an earlier age of onset of bipolar disorder, increased risk of suicide attempts, and co-occurring substance misuse (Aas et al., 2016; Park et al., 2020).

In addition, childhood trauma also influences the severity of bipolar symptoms a person experiences (Palmier-Claus et al., 2016; Rowland and Marwaha, 2018). In one literature review, researchers discovered that childhood maltreatment is associated with earlier onset of the disorder, more frequent and severe mood episodes, and increased risk of suicide (Palmier-Claus et al., 2016).

The number of adverse childhood experiences seems to be an important link to bipolar disorder as well. In a study of individuals with bipolar disorder in the U.S., Young-Min Park and colleagues (2020) found that an increased number of adverse childhood experiences was most strongly associated with negative outcomes rather than the type of adverse childhood experience. Among the adverse childhood experiences, physical abuse was most strongly linked with bipolar disorder, with 39.3% of individuals reporting at least one instance of physical abuse.



Poverty

Poverty is a chronic stressor that affects individuals world-wide. Poverty has specific physiological effects, including on children's brain development (e.g., Blair & Raver, 2016) and on a number of health markers in adults, including high blood pressure, obesity, cardiovascular disease, cancer, and respiratory disease (Brisson et al., 2020). **Toxic stress** is the excessive, prolonged and/or frequent activation of the stress response systems in the body and brain that increase the risk of health problems and stress-related illnesses. Toxic stress can occur when someone lives in persistent poverty or in a high poverty neighborhood with limited resources and increased exposure to negative environmental conditions (Brisson et al., 2020, p. 697; Shanks & Robinson, 2013). Social work researcher Daniel Brisson and colleagues (2020) note that toxic stress is the mediator between poverty and negative health outcomes in adults.

Based on research in the United States, Colleen Heflin and John Iceland (2009) found that all five forms of material hardship, including lack of medical care, free food, phone turned off, unstable housing, and problems paying bills, were linked with an increased risk of depression. In turn, they proposed that addressing any of these hardships may be one route to help decrease depression. In particular, the women who were most likely to be depressed were those who experienced problems paying bills (1.8 times more likely than without this hardship) and those who had their phone turned off (1.5 times more likely than those without this hardship).

Links between poverty and depression extend to children as well. In a study of 10 to 14 year old children in Turkey, Fikriye Yilmaz and her colleagues (2020) found that particular indicators of poverty were identified as significant risk factors for high depression rates. Those indicators included living in a single-parent family, rating health as poor, having any disease, deprivation according to the UNICEF Child Deprivation Index (CDI), going to bed or school hungry occasionally, and working to earn money.

While there is less research on the links between poverty and bipolar disorder, Michael Bauer and colleagues (2011) examined the relationship between median family income and mood symptoms in patients with bipolar disorder in the United States. Patients in the lower-income group spent fewer days euthymic and more days manic/hypomanic compared to patients in the middle- and upper-income groups. In this study there was no relationship between median income and depressive symptoms.



Prejudice and Discrimination

Creating categories of "us" and "them" is a natural and almost automatic process. It happens around trivial memberships (e.g., Coke vs. Pepsi) and less trivial memberships (e.g., Red Sox vs. Yankees). It happens with political parties (e.g., democratic vs. republican) and socioeconomic classes (e.g., top 1% vs. bottom 99%). It happens between countries (e.g., Ukraine vs. Russia). These distinctions affect our thoughts, feelings, and behaviors on a daily basis and are the basis for stereotypes, prejudice, and discrimination.

The generalizations we have about groups are known as **stereotypes,** and may be positive or negative. They are used to simplify and expedite perceptions and judgments and are resistant to change. Stereotypes may or may not include **prejudice,** which is defined as a negative attitude toward another person or group formed in advance of any experience with that person or group that is typically resistant to change. This is a simplification of Gordon Allport's original definition: "An antipathy based upon a faulty and inflexible generalization" (Allport, 1954, p. 9), which in our terms translates into "negative feelings based on beliefs (stereotypes)." Unlike stereotypes, prejudice is always negative. The behavioral component, **discrimination,** is differential treatment of the members of different racial, ethnic, religious, national, or other groups.

Discrimination is usually the behavioral manifestation of prejudice and involves negative, hostile, and injurious treatment of the people experiencing discrimination. For many individuals, experiencing discrimination is a chronic stressor that has been linked to poor health outcomes, including depression (Cuffee et al., 2012; Hart et al, 2021). Indeed, one meta-analysis examined the effects of discrimination broadly, including based on race, ethnicity, national origin, religion, gender, sexual orientation, disability, and weight and concluded that the perception of discrimination had harmful mental health outcomes, including depression (Schmitt et al., 2014).

Racial Discrimination

Racism is a form of prejudice that assumes that the members of racial categories have distinctive characteristics and that these differences result in some racial groups being inferior to others. Racism generally includes negative emotional reactions to members of the group, acceptance of negative stereotypes, and racial discrimination against individuals; in some cases it leads to violence. There is increasing awareness of the longstanding link between experiencing racism and depression. In fact, dozens of studies have examined the link between discrimination based on ethnoracial groups and depression. One study examined the association between everyday discrimination and depression in Black women in the U.S. at two points in time over a five year period. The participants were asked questions such as: "How often have other people acted as if they were better than you?" The researchers found that everyday discrimination and depression were significantly correlated; increased discrimination was associated with increased depression (Schulz et al., 2006).

Psychologist Angelitta Britt-Spells and her colleagues reviewed research examining the effects of perceived discrimination on depressive symptoms among Black men in the U.S. (Britt-Spells et al., 2018). Across twelve separate studies, greater perceived discrimination was significantly correlated with an increase in depressive symptoms (see also Ward & Mengesha, 2013; Watkins et al., 2011). Research has found a significant association between racial discrimination and depression to confidently add depression to the list of negative outcomes of racism (Carter et al., 2019; Pieterse et al., 2012).

David Williams and Serena Mohammad (2009) noted that discrimination and racial disparities in healthcare can exacerbate psychopathology both by increasing the likelihood of people developing a disorder and by limiting people's ability to access care for the disorder. Others have noted harmful experiences faced by members of marginalized groups. For example, Psychologist Kira Hudson Banks and her colleagues (2006) examined the influence of perceived everyday discrimination on mental health outcomes in a survey of Black Detroit residents. The researchers used data from the 1995 Detroit Area Study Discrimination Questionnaire (DAS-DQ; Jackson & Williams, 2002), which asked respondents how often they had experienced everyday discrimination, such as not being treated with as much courtesy as others and having others behave as if they are better than you. The researchers then measured the participants' symptoms of depression and anxiety and found that increased perceived everyday discrimination was associated with increased levels of both depression and anxiety (see also Hammond, 2012).

Clinical psychologist Tina Chou and her colleagues used the same measure of discrimination (Jackson & Williams, 2002) to compare the effects of everyday discrimination on mental health outcomes of Black, Asian, and Latinx Americans (Chou et al., 2012). Chou and her colleagues measured the prevalence of disorders in five categories: anxiety disorders, mood disorders, eating disorders, substance use disorders and other disorders. They reported, first, that Black Americans indicated they had experienced a higher degree of racial discrimination than Asian and Latinx Americans. Second, across all three groups, racial discrimination was associated with poor mental health outcomes. Specifically, Black Americans were more likely than Asian Americans to report anxiety disorders, and Latinx Americans were more likely than Asian Americans to report anxiety disorders and depression, and more likely than Black Americans to report depression. Chou and colleagues suggested that these differences may, in part, be due to differences in the sociopolitical history of the three groups. In the United States, Black Americans have a much longer history than Hispanic or Asian Americans of battling racism and discrimination that likely causes them to be more attuned to and vulnerable to instances of discrimination.

This link between racial discrimination and depression remains strong outside of the U.S. as well. Chou (2012) found that new migrants to Hong Kong who reported discrimination were more likely to have increases in depression one year later. Similarly, a population based study of 23 European countries found that immigrant and ethnic minority groups had a higher rate of depressive symptoms compared to native residents (Missinne & Bracke, 2012).

In sum, research over the last thirty years has supported the conclusion that experiencing racism is associated with poorer mental health, including increased rates of depression, anxiety, and psychological stress (Paradies et al., 2015). Further, researchers have identified stress caused by racism as one mechanism by which racial discrimination predicts negative psychological and physical health outcomes (Neville et al., 2004; Williams & Mohammed, 2009). It is important that the focus of ongoing research include both reducing the stress caused by prejudice and discrimination and on treating the associated mental health outcomes. More importantly, as societies around the world work to address systemic racism, they must recognize its role in mental health.

Gender and Sexuality Discrimination

A large and still growing body of research shows that the way individuals are treated in society based on their gender identity and sexual orientation has implications for their mental health. For example, a study of over 4,500 Czech women who were followed over approximately 11 years beginning in mid-pregnancy found that participants who had experienced gender discrimination were significantly more likely to develop depression than those who had not (Stepanikova et al., 2020). Additionally, violence against women is a serious, global social problem with significant mental health consequences (Ellsberg et al., 2008; Kumar et al., 2013; Oram et al., 2017). Epidemiologist Mary Ellsberg and her colleagues examined data from the World Health Organization's intimate partner violence survey of more than 24,000 women from ten countries. Women who reported at least one incident of intimate partner violence experienced more emotional distress, suicidal ideation, and suicide attempts than women who did not (Ellsberg et al., 2008).

Targets of sexuality discrimination are often members of the LGBTQ+ community. The initials "LGBTQ" is used to refer to members of a culture that includes people who are lesbian, gay, bisexual, transgender, queer, and/or questioning their sexual identity or orientation. Because so many different groups are brought under a single umbrella term, it's important to note that there is not just one LGBTQ culture, but rather a multitude of cultures made by the intersection of these many categories. For example, the lived experiences and norms of a gay transgender man are likely very different from those of a cisgender bisexual woman. To clarify some terminology, a person's gender identity is their self-identification as male or female, both, or nonbinary, among others. The term cisgender indicates a person's gender identity matches with their sex assigned at birth, whereas the term transgender indicates the two do not match. This distinction is separate from a person's sexual orientation—the term that indicates who an individual is sexually attracted to.

In the U.S., lifetime prevalence rates for experiencing any mental illness are twice as high among the LGBTQ population compared to the heterosexual, cisgender population (Kates et al., 2018). Further, according to measurements from U.S. and U.K. populations, LGBTQ individuals are up to 2.5 to 3 times more likely to experience depression or anxiety (Cochran et al., 2003; Hatzenbuehler et al, 2008; Semlyen et al., 2016), and they are more likely to attempt suicide (Hatzenbuehler, 2011).

These associations are also evident early in development. A sample of Dutch youth from 11 to 22 years of age found that LGB youth experienced higher rates of depressive symptoms compared to heterosexual youth. When separated by gender, LB girls at 11 years of age already experienced increased rates of depressive symptoms, which increased over time (la Roi et al., 2016).

Why are LGBTQ individuals more likely to develop psychological disorders than heterosexual, cisgender individuals? Recall that researchers identified stress resulting from being the target of racism and discrimination as an underlying factor in psychological disorders. Members of the LGBTQ population, especially adolescents, also tend to experience prejudice and discrimination from their local community, their peers, and their families.

Psychologist Ilan Meyer (2003) reviewed the literature examining sexual orientation and mental health outcomes and suggested that LGBTQ individuals experience distress based on their minority status that is different from general psychological stress. Clinical psychologist Mark Hatzenbuehler (2009) proposed a model in which stigma-related stress leads to negative mental health outcomes (e.g., increased depression, anxiety, substance use) through three processes: rumination, social isolation, and hopelessness/pessimism. LGBTQ individuals who experience stigma-related stress that leads to rumination, social isolation, and feelings of hopelessness and pessimism will likely experience increased depression, anxiety, and substance use.

Intersectionality and Discrimination

Many researchers have also reported on various intersectionalities and depression. Intersectionality examines the impact of multiple underrepresented social identities present in one individual on outcomes of interest, such as depression. For example, LGBTQ individuals who are also part of an underrepresented racial group (e.g., Black individuals) experience more discrimination, with its mental health implications, than their White counterparts (Whitfield et al., 2014).

Roberto Abreu and colleagues (2023) found that youth who were Latinx and transgender reported more intersectional microaggressions (e.g., instances of People of Color Heterosexism, LGBT Racism, and LGBT Relational Racism) and more depressive symptoms than cisgender peers. Likewise, another study that examined the role of intersectional stressors found that for sexual and gender minority adolescents of color intersectional stressors were associated with more depressive symptoms than racism or sexual and gender minority-based discrimination alone (Mereish et al., 2022). Continued research on the intersectionality of race, sexual orientation, and gender, and its influence on the development of depression is needed to understand the complexity of this area.